

5-Mark Questions & Answers — DBMS Module 4: Relational Database Design

Q1. Explain 1NF, 2NF, 3NF, and BCNF with concrete table examples.

1. **1NF (Atomic Values):** Eliminates multi-valued cells or comma-separated lists.
2. **2NF (Full Functional Dependency):** Removes partial key dependencies ($A \rightarrow B$ where A is part of key {A, C}).
- Fix: Split into $R1(A, B)$ and $R2(A, C)$.
3. **3NF (No Transitive Dependency):** Removes $X \rightarrow Y$ where neither X is superkey nor Y is prime.
- Fix: Move $X \rightarrow Y$ to a separate lookup table.
4. **BCNF (Strict Super Key):** Forces every determinant X in $X \rightarrow Y$ to be a super key.

Q2. Explain the 3NF synthesis algorithm for lossless-join and dependency-preserving decomposition.

1. Compute Minimal Cover F_c for functional dependency set F.
2. For each FD $(X \rightarrow Y)$ in F_c :
Create a new relation schema $R_i = X \cup Y$.
3. If no schema R_i contains a Candidate Key of original relation R:
Create an additional schema containing any Candidate Key of R.
4. Remove redundant sub-schemas (if R_i is a subset of R_j , delete R_i).

- **Guarantee:** Resulting schemas are in 3NF, preserve all functional dependencies, and guarantee lossless join.

Q3. Explain 4NF and decomposition using multivalued attributes with an example.

- **4NF Condition:** A BCNF relation is in 4NF if for every non-trivial MVD $X \twoheadrightarrow Y$, X is a super key.
- **Example Violation:** $STUDENT(ID, Mobile_No, Skill)$ where Mobile_No and Skill are independent multi-valued attributes.
- **Decomposition:**
 1. $STUDENT_MOBILE(ID, Mobile_No)$
 2. $STUDENT_SKILL(ID, Skill)$

Q4. Describe NoSQL Data Models (Key-Value, Document, Column-Family, Graph) with use cases and CAP theorem.

- **CAP Theorem:** Distributed systems can achieve at most 2 out of 3: **Consistency, Availability, Partition Tolerance**.
- **Data Models:**
 1. **Key-Value:** Caching, Session tokens (Redis).
 2. **Document:** JSON-based content catalogs (MongoDB).
 3. **Column-Family:** High-throughput time-series analytics (Cassandra).
 4. **Graph:** Network relationships & social graphs (Neo4j).

